Report on Waterfall Model

The Satelite on the Research of the software Life Cycle: V+ Iterative Waterfall

Abstract:

This report presents a case study that explores the implementation of the Waterfall model in The Satelite on the Research of the software Life Cycle: V+ Iterative Waterfall Project.The Waterfall model is a traditional and sequential approach to project management, involving distinct phases. The development of data process system in a preresearch subject is challenge and explorative. Obvious, the traditional life cycle of Waterfall mode is not applicative to the process of the software. To resolve this problem, a V + iterative + waterfall model is applied to the development of the pre-research subject EP satellite data process system by taking into account the character of the project Team and the software .

Introduction:

The Waterfall model is one of the earliest and most well-known software development methodologies in the field of software engineering. It follows a linear and sequential approach to project management, where each phase must be completed before moving on to the next one. The model is called "Waterfall" because the progress flows in one direction, like a waterfall, and it is challenging to go back to previous phases once they are completed. This project highlights the advantages of space-based observations over ground-based ones due to the absence of Earth's atmosphere, which allows for better quality data. It also mentions the increased types and quantities of payloads in space projects and the need for effective software development and quality assurance activities to meet the scientific goals of astronomy missions.

Problem Statement:

The traditional waterfall model, though widely used in space software development, is criticized for its lack of feedback and limited communication between stakeholders. Therefore, a combination of the waterfall model, V model, and iterative model is proposed to create a more suitable approach for space pre-research projects. The V++ waterfall + iterative model is suggested as a more reasonable, scientific, and effective solution to guide the software development process for space missions.

The Application and Implementation of Waterfall model on the problem:

Waterfall model, also known as linear order model, it provides a top-down, structured software development approach. And it is very famous for its definition of the lifecycle. The activities of the software life cycle in a fixed order execution: demand, design, construction, integration, testing, maintenance, there are clear boundaries between the various activities and no continuous, the end of the last stage of the results through the verification or review to start the next stage of the work. The activities are independent of each other, not overlap, and its shape like a waterfall, so called waterfall model. The main stages of the waterfall model are as follows: planning, requirement analysis, software design, programming, software testing, operation and maintenance, etc. The waterfall model is the basis of all the software lifecycle models. The applications are as follows:

(1) The user's requirements are very clear, and have been well understood, the documents are very clear.

(2) The product of the project has a clear definition, and a technical solution that is easy to understand.

(3) The user's requirement for product quality is higher than the requirement of cost and schedule, the waterfall type

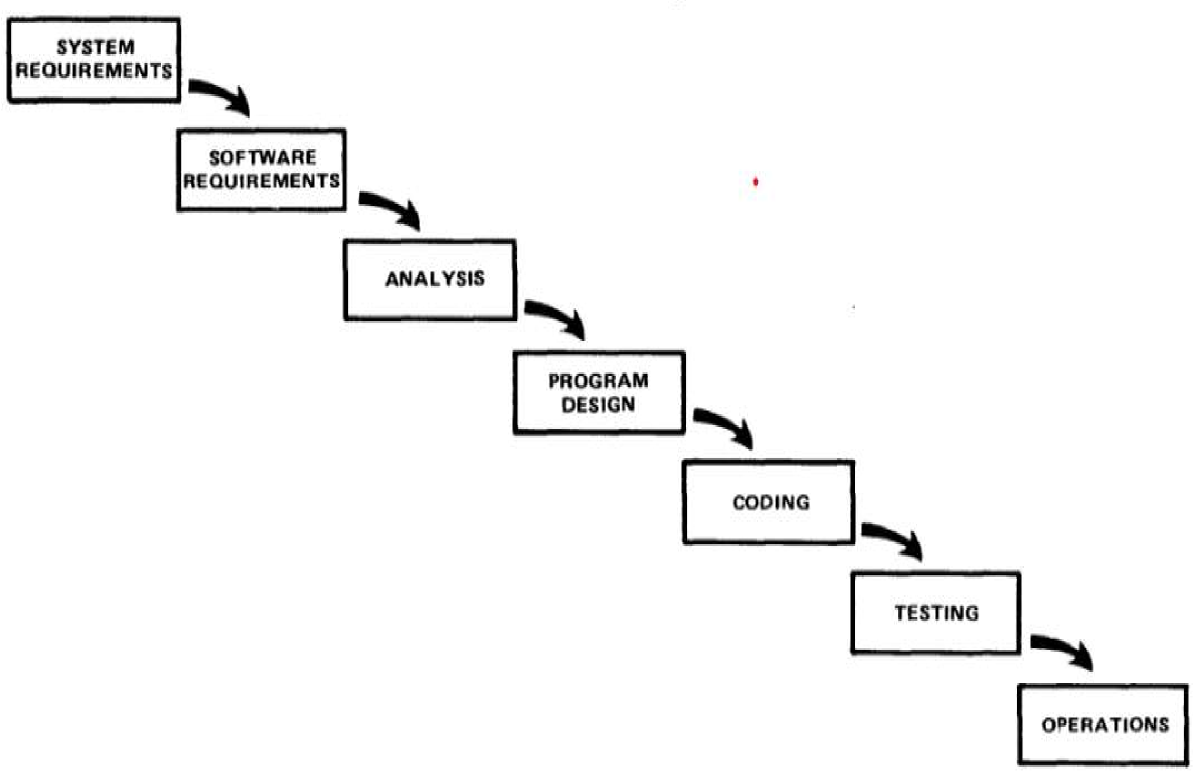

Fig.1. The definition of Waterfall model and Application

The implementation of each stage is shown in the above table, that is to say, demand stage review, received user feedback, needs refinement; design, unit testing, phase feedback and iteration, and the corresponding test plan, improve the unit test plans, system integration test plan, test plan; unit test, regression test the test case, usually placed on the description of the test, improve the integration test plan; integration test, regression test, improve the integrated configuration test; configuration test, regression test, improve the test system .

The research paper describes the stages of the study, which consisted of requirement gathering and analysis, design, implementation, testing, deployment and maintenance.

- **Requirements analysis and technical requirements in the pre-research stage:**

According to the scientific goals proposed by the user, here we simply analyze the requirements and form the functions that the software needs to complete.
The technology requirements as follows:
(1) Based on the received X ray photon incident flow, the X ray source is extracted and the location of the X ray source is calculated
(2) Analysis of X ray source, through the comparison with the known catalog.
(3) Total processing time is less than 2 minutes

function	*requirement*
accept	acceptance of X ray photon flow
extract	extraction of X - ray signals higher than a certain signal to noise ratio
calibrate	flow calibration and space coordinate conversion
Compare star catalog	The source of the extraction is compared with the known star Mark, classification: the known source and the new source
trigger	Sending trigger alerts for scientific targets

- **Design :**

The change of the user's requirements design integration test cases Coding Code modification , design integration test cases design unit test cases Coding Code modification

- **Implementation :**

The implementation of each stage is shown in Table 1, that is to say, demand stage review, received user feedback, needs refinement; design, unit testing, phase feedback and iteration, and the corresponding test plan, improve the unit test plans, system integration test plan, test plan; unit test, regression test the test case, usually placed on the description of the test, improve the integration test plan; integration test, regression test, improve the integrated configuration test; configuration test, regression test, improve the test system Table 1

Stage	Activity	Export
Requirements analysis	communicate with user Design functional test cases	External review
Outline design	The change of the user's requirements Design integration test cases Coding Code modification	Internal review
Detailed design	Design integration test cases Design unit test cases Coding Code modification	Internal review
Coding and unit test	Execute unit test Code modification Integration case modification	Internal review
Integration test	Execute integration test Code modification Configuration item test modification	Internal review
Configuration item test	Execute Configuration item test Code modification System test cases modification	Internal review
System test	Execute system test Code modification regression testing	External review

Table. I. Implementation list of each stage

- **Testing :**
  Execute Configuration item test code modification system test cases modification. Execute unit test Code modification Integration case modification. Execute integration test Code modification Configuration item test modification. Configuration item test Execute Code modification cases modification System test Execute system test Code modification regression testing. The feedback of the testing case is asked to be focus on. For the Pre-research project, the data process software of the EP satellite, the test is asked to be focused on all the stage. The feedback of the results was needed to communicate with the user to ensure agree with the scientist goal of the users.

- **Deployment:**
  The data process software of the EP satellite, the test is asked to be focused on all the stage. The feedback of the results was needed to communicate with the user to ensure agree with the scientist goal of the users.

- **Maintenance**:
  Repair process of the system as agreed.

Result :

The chosen of the software life-cycle should to consider the characters of the project. The requirements can be refined, the test can be verified, and the requirement of each stage can be clear. Therefore, the process of each stage divided can be worked on the development. The actual satellite borne trigger and the star ground data processing need to be experimentation and software engineering on the actual computer unit on the satellite. The software of satellite data processing is mainly based on the simulation the data processing process on the Star software simulator, which belongs to the background type pre research project, and the basic algorithm has been verified through feasibility.To resolve this problem, a V + iterative + waterfall model is applied to the development of the pre-research subject EP satellite data process system by taking into account the character of the project Team and the software . The results show this method can supply the guidance for the software developers.

Conclusion:

In this paper, we clarify the new life-cycle is chosen in a pre-research satellite project. A adopt software life-cycle is the first step for a software development, which makes the process more easier, more control, and reduce the risk. For the preresearch satellite project, the test of the software and the feedback from the user are through all the stage of the development, a V model-Iteration –waterfall model is adopted.

The Satelite on the Research of the software Life Cycle: V+ Iterative Waterfall

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***Abstract*—The development of data process system in a pre-research subject is challenge and explorative. Obvious, the traditional life cycle of Waterfall mode is not applicative to the process of the software. To resolve this problem, a V + iterative + waterfall model is applied to the development of the pre-research subject EP satellite data process system by taking into account the character of the project Team and the software . The results show this method can supply the guidance for the software developers.**

Keywords—Satellite Pre-research; life—cycle;Data Process

I. INTRODUCTION

The information of the whole band can be observed, due to the absence of the earth's atmosphere of the outer space of the earth, which is becoming the favorite field for the astronomer to obtain the celestial body of physical information. With the development of space astronomy technology, more and more astronomical instruments have been sent to space to meet Astronomers' scientific requirement which has promoted the types and quantities of payloads in space projects. In facts, the data of space observation are indeed not comparable with those of the ground instruments. Because no influenced by the atmosphere outer the earth, the data observed by the space telescope are better than the ground telescope with the same aperture. For example, HST, the data is provided bring a lot of now processes. In China, the data from the optical payload lunar optical telescope launched on in China prove that there is no water on the surface of the moon [1], the result is the best in recent years. Some researches of the variable stars are also published [2,3].

With the development of information technology, software product is not only an independent product, but also a decisive product. Many of the pre research projects are often challenged by their software designed, especially for the space Pre-research project. There are no experience and no real environment to be provided. For the space pre - research project of the satellite background model, the software needs to meet the requirements of the scientific goal of the astronomy, at the same time it requires high reliability and security [4], to ensure the smooth implementation of the pre research task to the maximum extent and finished. Therefore, effective software phase division and quality assurance activities must be adopted to guide developers to develop high-quality software and meet mission objectives in the process of software development.

The waterfall model is very popular during the development of the space software, for its character with a strict definition of the stage and easy to control. However, the traditional waterfall model is usually on a stage is the next phase of the input, and there is little feedback in each stage of the development process. It is often to the configuration of test stage to see the whole development process of the results, which cannot be reflected the communication between people software tools and links [5]. Facing the various stages of development process, the satellite projects for pro-research that may be needed to be tested and returned the results during the developing, the waterfall model seems to be somewhat inappropriate. A V model intended the water falling model is mostly used in many test [6]. Even the test model is based on the V model to be used [7]. For the management of space astronomy software developing process, an iterative model is proposed [8]. Because of its innovativeness characteristics of the space Pre-research project, obviously, a single waterfall model or iteration model are not applicable.

In this paper, we study a life-cycle model that is used in the development of the pre-research project: Einstein probe data Process. Based on the background of software testing throughout the process of software development, mainly adopts the V model, we study to combine the waterfall model, V model and iterative model during the software life-cycle, and to make the software development stage more reasonable, scientific and effective. The structure is organized as follows: Section 2 presents the instruction of the waterfall model, V model and the iterative model. Section 2 the application of V++ waterfall + iterative model. Section 3 the character of the pre-research introduction of the satellite project. Section 4 concludes the paper.

II. THE INSTRUCTION OF THE WATERFALL MODEL, V MODEL AND THE ITERATIVE MODEL.

A. *The Application and Problems of Waterfall Model*

Waterfall model, also known as linear order model, it provides a top-down, structured software development approach. And it is very famous for its definition of the life-cycle. The activities of the software life cycle in a fixed order execution: demand, design, construction, integration, testing, maintenance, there are clear boundaries between the various activities and no continuous, the end of the last stage of the results through the verification or review to start the next stage of the work. At the end of each activity should be reviewed and to be determined as the beginning of the next phase of activities. The activities are independent of each other, not overlap, and its shape like a waterfall, so called waterfall model. The main stages of the waterfall model are as follows: planning, requirement analysis, software design, programming, software testing, operation and maintenance, etc. The waterfall model is the basis of all the software lifecycle models. Actually, the waterfall was suggested in a deal institution in the origin paper [9].

The waterfall model is applied to each phase strict defined clearly. Therefore, the main work products can be entered into the next stage by testing or reviewing, and the completion of every activity is marked by the evaluation of the activity for this strict phased requirements.

The applications are as follows:

(1) The user's requirements are very clear, and have been well understood, the documents are very clear.

(2) The product of the project has a clear definition, and a technical solution that is easy to understand.

(3) The user's requirement for product quality is higher than the requirement of cost and schedule, the waterfall type model is applicable. By using this model, it helps the developer to identify the problem early and reduce the stage cost.

However, in the practice of software development research projects, usually the user's requirement is not clear. The scientists are custom to express the scientific goals that they hope to observe, and the software developers transfer the requirements to the software can achieve some of the functions during the developing. Even some function is through the development and testing of continuous clear process. A part of requirements are clear in the development process. All of these characters of the process, which lead to the difficulty and complexity of software development technology. Therefore, it is very obvious that the waterfall model is difficult to adapt to the development of software project with the characteristics of uncertain demand and complex technology during process of the development software.

B. The Application and Problems of V Model

The V model is a continuous development model, very similar to the waterfall model. The V models can be generally divided into the following stages: demand analysis, outline design, detailed design, software coding, unit testing, integration testing, configuration item testing, system testing, acceptance testing.

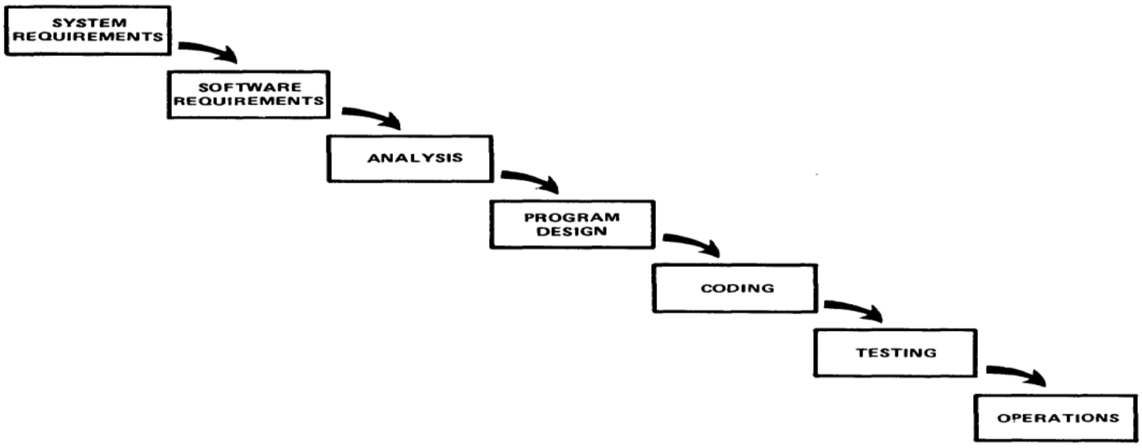

Fig. 1. The defination of waterfall model in the earlier application

The difference from the waterfall model is that there is a test stage matching each development stage: requirement stage is the same as system test stage, architecture design stage is integrated test stage, subsystem design stage is same as the unit test stage. The V model is an improvement of the waterfall model.

In the V model, the general unit test corresponding to the detailed design of the link, that is to say, the software testers with the development people need to out together in the detailed design of test cases for unit testing. The integrated testing is matched with the outline design, that mean, the integration test cases are be designed according to realize the method of the outline design and function module interface, in the analysis and functional analysis of interface module, data transmission method, and prepared for the use of integration test . The system testing is based on the demand analysis and system analysis. When the system developer are preparing the system analyses according to the requirements specifications, the test personnel should write out all kinds of test cases can finally realize the function of the system according to the requirements specification, to prepare for the final system test.

The model emphasizes the corresponding relationship between the test stage or process and the development process. From the model, the software testing is the focus of the V model. And the software testing activities are relative to each stage of development activities.

The applications are as follows:

(1) Stable and reliable requirements

(2) The method of software development is very mature

(3) The software structure is clear and the boundaries between modules are very clear.

The V model helps to delay software projects in the avoidance and design stages that may cause problems. However, the V model only takes the test process as a stage after requirement analysis, system design and coding. It ignores the verification of test for requirement analysis and system design. That leads to that the users' needs are verified until late acceptance test. And it's not enough to deal with risk. If there are significant changes in project requirements, it is necessary to go to the previous stage's framework, design, code or test related documents, which will cost a lot. At the same time, the software product is not visible to the user during the development process, and it also leads to the insufficient participation of the users during the development process.

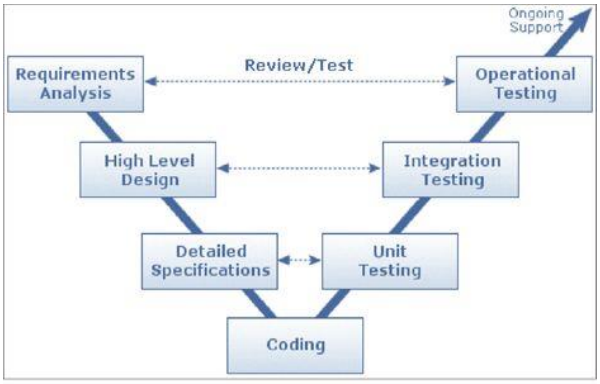

Fig. 2. Software development life-cycle:V model

C. *The Iterative development*

Iterative and progressive development is the basis of modern software methods [10]. The earliest iterative process may be described as stagewise model, used US Air Force SAGE project lead by H.D.Benington. The rational Unified Process choses the model during the development [11], and is a duplicated and iterative software development process. It is impossible to define all the problems in order, design a complete solution, compile software, and test the product at the end because of the process of the modern software system turns out to be more complex. Through repeatable iterations to understand, problems are solved effectively on the basis of multiple iterations. RUP through the software development cycle is divided into the initial delivery of refinement, structure and four stages, each stage contains several iterations, and each of iteration is equivalent to a mini waterfall model, including requirements analysis, design, coding and testing, each of iteration is a previous iteration as the foundation.

In iterative development, development work can be completed before the requirements are determined， ench of the iteration complete part of the system function. At the same time the users can feedback and refine their requirements or add new requirements, then a new iteration start.

The applications are as follows:

(1) Can reduce the risk

(2) Can get early feedback from the users

(3) Continuous testing and integration

(4) Speed up the progress of development

III. THE APPLICATION OF V+WAGTERFALL+ ITERATIVE MODEL

A. *Project introduction*

The Einstein probe project is supported by the strategic pilot science and technology project of the Chinese Academy of Sciences. In scientist, it is to resolve the following three aspects of Astrophysics and basic physical questions:

(1) To find and observe different kind of black hole

(2) Gravitational wave

(3) X-ray temporary source

Because of the space project, the actual satellite borne trigger and the star ground data processing need to be experimentation and software engineering on the actual computer unit on the satellite. The software of satellite data processing is mainly based on the simulation the data processing process on the Star software simulator, which belongs to the background type pre research project, and the basic algorithm has been verified through feasibility.

B. *Requirements analysis and technical requirements in the pre-research stage*

According to the scientific goals proposed by the user, here we simply analyze the requirements and form the functions that the software needs to complete.

TABLE I. LIST OF SCITICENT REQUIREMENTS ANALYSIS

function	*requirement*
accept	acceptance of X ray photon flow
extract	extraction of X - ray signals higher than a certain signal to noise ratio
calibrate	flow calibration and space coordinate conversion
Compare star catalog	The source of the extraction is compared with the known star Mark, classification: the known source and the new source
trigger	Sending trigger alerts for scientific targets

The technology requirements as follows:

(1) Based on the received X ray photon incident flow, the X ray source is extracted and the location of the X ray source is calculated

(2) Analysis of X ray source, through the comparison with the known catalog.

(3) Total processing time is less than 2 minutes

The requirements of achievement as follows: For the source extract and accept, it needs to do background assessment, distinguish source type. The extraction algorithms of scientist target need to be tested to ensure. For the requirement 2,3, they are decided and proved by testing repeat.

In this way, we summarize the development characteristics of data processing software on EP satellite as follows:

(1) The degree of software engineering is very high, and the development stage must be clear

(2) The part of the requirements are clear, another not

(3) There is new and unpredictability in the development process

(4) Continuous test validation runs through the entire development process

C. *The chose of the lify cycle*

Considering the basic algorithm of the EP satellite data processing software has been validated, we decided using the framework of V model. By the way, something like the V model in the test to verify the neglected requirements and design, risk control and lack of user participation is insufficient, is also taking into account. So we adopted on the stage the waterfall model must be carried out review and verify, input and output of each phase. The user feedback is gotten and realized by an iterative approach repeat. The refinement requirements are worked through very stage.

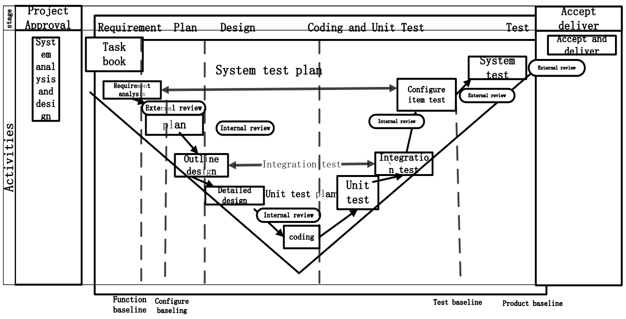

Fig. 3. Improvement V model l

The implementation of each stage is shown in Table 2, that is to say, demand stage review, received user feedback, needs refinement; design, unit testing, phase feedback and iteration, and the corresponding test plan, improve the unit test plans, system integration test plan, test plan; unit test, regression test the test case, usually placed on the description of the test, improve the integration test plan; integration test, regression test, improve the integrated configuration test; configuration test, regression test, improve the test system Table 2.

TABLE II. IMPLEMENTATION LIST OF EACH STAGE

stage	*activity*	*export*
Requirements analysis	communicate with user Design functional test cases	External review
Outline design	The change of the user's requirements Design integration test cases Coding Code modification	Internal review
Detailed design	Design integration test cases Design unit test cases Coding Code modification	Internal review
Coding and unit test	Execute unit test Code modification Integration case modification	Internal review
Integration test	Execute integration test Code modification Configuration item test modification	Internal review
Configuration item test	Execute Configuration item test Code modification System test cases modification	Internal review
System test	Execute system test Code modification regression testing	External review

The proportion of the elements that affect the elements is also different at every stages of implementation.

(1) Requirement stage: It is important to obtain and understand the requirements of the user. The communication is emphasized with the users. And the feedback for the function of the user's is also needed.

(2) Outline stage and detailed stage: The changes of the requirements become smaller. The feedback of the test cases and realize of the coding is emphasized.

(3) Test stage: the feedback of the testing case is asked to be focus on.

For the Pre-research project, the data process software of the EP satellite, the test is asked to be focused on all the stage. The feedback of the results was needed to communicate with the user to ensure agree with the scientist goal of the users.

IV. CONCLUSIONS

In this paper, we clarify the new life-cycle is chosen in a pre-research satellite project. A adopt software life-cycle is the first step for a software development, which makes the process more easier, more control, and reduce the risk. The feedback through all the stage is made clear the requirements of the user's step by step. The chosen of the software life-cycle should to consider the characters of the project. For the pre-research satellite project, the test of the software and the feedback from the user are through all the stage of the development, a V model- Iteration –waterfall model is adopted. The requirements can be refined, the test can be verified, and the requirement of each stage can be clear. Therefore, the process of each stage divided can be worked on the development.

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